Luminance and Contrast equalisation

Contrast definition:

We use the Root mean square (RMS) equation to define contrast.

(From Wikipedia)
"RMS contrast is defined as the standard deviation of the pixel intensities."

Intensity is referred as the Luminance (luminance is defined in next page)

The contrast formula of an image is:

$$\sqrt{\frac{1}{MN}\sum_{i=0}^{N-1}\sum_{j=0}^{M-1}(I_{ij}-\bar{I})^2},$$

I = the pixel intensity at position i,j
$\bar{I}$ = the intensity mean of all pixels
N the pixel row size
M the pixel column size

Luminance definition:

We use the **sRGB** standard that uses **ITU-R Recommendation BT.709** wich gives the conversion from a pixel to the luminance of that pixel.

From Wikipedia: "*sRGB uses the ITU-R BT.709 primaries, the same as are used in studio monitors and HDTV, and a transfer function (gamma curve) typical of CRTs. This specification allowed sRGB to be directly displayed on typical CRT monitors of the time, a factor which greatly aided its acceptance*"

The conversion is:

Y = a * Red + b * Green + c * Blue.

Y is Luminance and is between 0 and 1 inclusive.
a,b,c are the coefficients to convert from RBG to Luminance (Y)
a = 0.2126
b = 0.7152
c = 0.0722

To convert a pixel RGB color to Luminance and vice versa, we use the YUV color space.

The conversion matrices from RGB to YUV and vice versa are:

$$\begin{bmatrix} Y' \\ U \\ V \end{bmatrix} = \begin{bmatrix} 0.2126 & 0.7152 & 0.0722 \\ -0.09991 & -0.33609 & 0.436 \\ 0.615 & -0.55861 & -0.05639 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1.28033 \\ 1 & -0.21482 & -0.38059 \\ 1 & 2.12798 & 0 \end{bmatrix} \begin{bmatrix} Y' \\ U \\ V \end{bmatrix}$$

Luminance equalisation formula

luminance = strength * (luminanceToEqualise / luminanceMean) * luminance;

$$Luminance_i' = strength * (\frac{\overline{LuminanceMeanAllImages}}{LuminanceMeanImage}) * luminance_i$$

Luminance i = luminance of pixel at point i
Strength is a parameter controlling all equation, 0 meaning no equalisation and 1 being total equalisation.

Contrast equalisation

luminance = luminanceAllMean + strength * ((contrastAllMean / contrastMean) * (luminance - luminanceAllMean))

$$Luminance_i' = \overline{LuminanceMeanAllImages} + strength * (\frac{\overline{RMS\ Contrast\ all\ images}}{RMS\ Contrast\ image}) * \left(Luminance_i - \overline{LuminanceMeanAllImages}\right))$$

Luminance i = luminance of pixel at point i
Strength is a parameter controlling all equation.
0 meaning no equalisation and 1 being total equalisation.